

Quiz Question

Time: 20 minutes

Marks: 20

Scenario. Students of **Class A** often come to class mainly to ensure their attendance is counted. To automate attendance, the department designs a CNN named **AttendanceNet**, which recognizes one of 10 student IDs from a face image.

AttendanceNet Architecture

Layer	Specification
Input	$64 \times 64 \times 3$
Conv1	16 filters, $F = 3$, $S = 1$, $P = 1$
MaxPool1	2×2 , $S = 2$
Conv2	32 filters, $F = 3$, $S = 1$, $P = 1$
Conv3	32 filters, $F = 3$, $S = 1$, $P = 1$
MaxPool2	2×2 , $S = 2$
Conv4	64 filters, $F = 3$, $S = 1$, $P = 1$
Flatten	–
Dense1	128 neurons
Output	10 neurons

1. [4 marks] Find the tensor size just before the Flatten layer.
2. [3 marks] Calculate the number of values produced by the Flatten layer.
3. [4 marks] Calculate the number of trainable parameters in Conv2 and Conv3. Explain why they are not necessarily the same even though both use 32 filters.
4. [4 marks] Calculate the number of trainable parameters in Dense1. Based on your answer, explain why Dense1 may dominate the total parameter count.
5. [3 marks] Suppose the department wants to reduce the model size without changing the convolution layers. Suggest one architectural change after Conv4 and explain why it reduces parameters.
6. [2 marks] A student claims: “MaxPool2 reduces the depth from 32 to 16.” Is the claim correct? Justify briefly.